Understanding User Preferences through Latent Factors

Evaluating Matrix Factorization vs. Neighborhood-Based Collaborative Filtering on MovieLens 100K

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The Data Science Challenge: Information Overload

Streaming platforms curate massive catalogs, creating severe **information overload** for end consumers.

- **Key Catalyst:** Over 80% of all video plays are driven directly by recommended pipelines on major networks.
- **Scientific Inquiry:** Can low-rank latent factor representations outmatch standard localized neighbor metrics?
- **Hypothesis:** Matrix Factorization maps sparsity (93.7%) better by compressing profiles into robust lower-dimensional spaces.

Data Overview & Project Pipeline

MovieLens 100K Profile

Parameter	Value / Attribute
Total Audited Users	943 Active Consumers
Catalog Size	1,682 Movies Index
Observations	100,000 Verified Explicit Ratings
Sparsity Index	**93.7% Sparse Matrix**
Evaluation Split	u1.base (80k Train) / u1.test (20k Test)

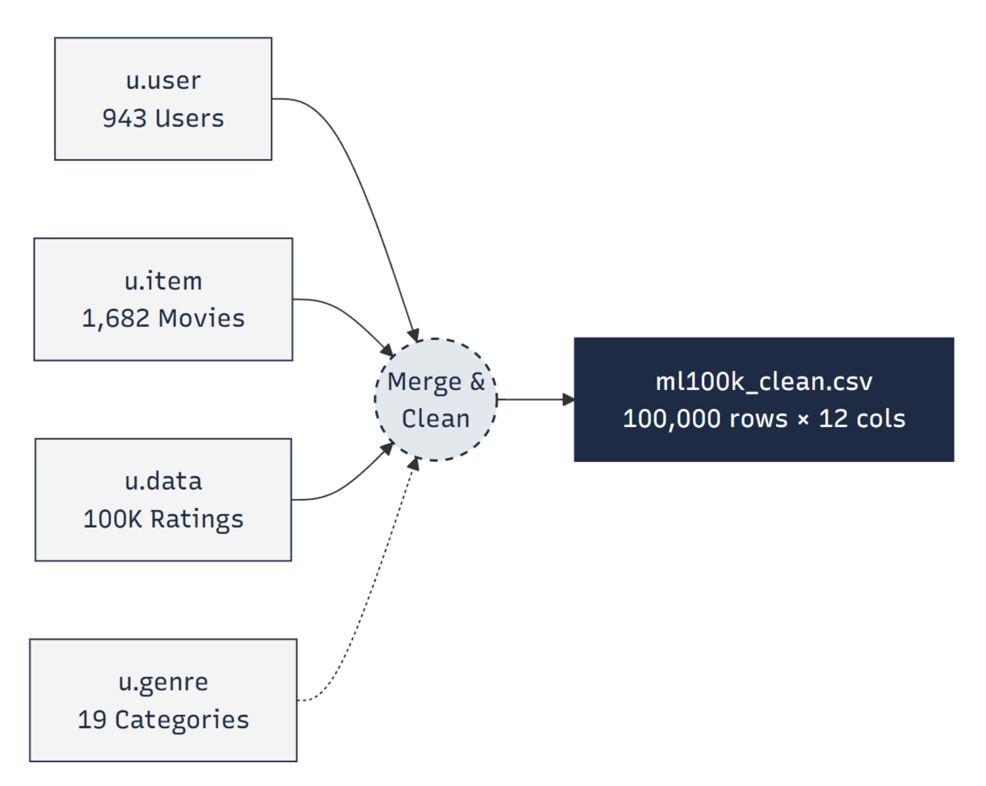
Execution Timeline

The pipeline strictly adhered to standard collaborative systems engineering steps from acquisition to benchmarking.

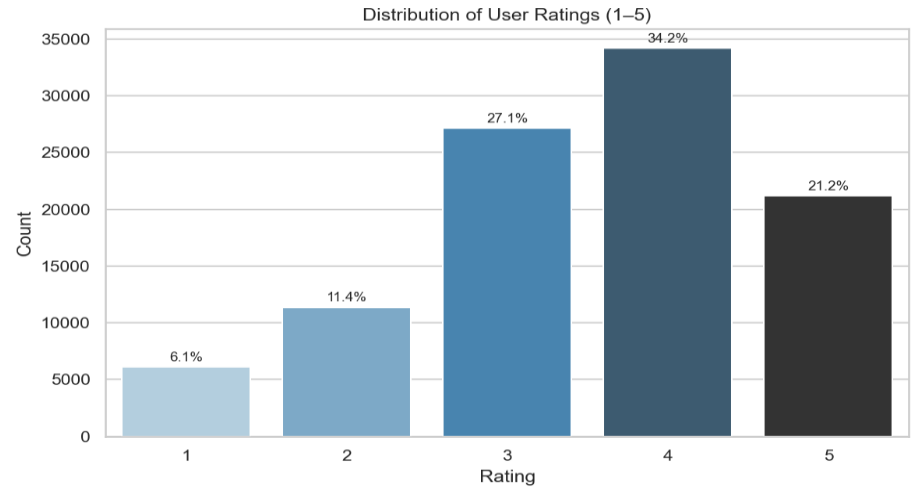
From Raw Data to Clean Dataset

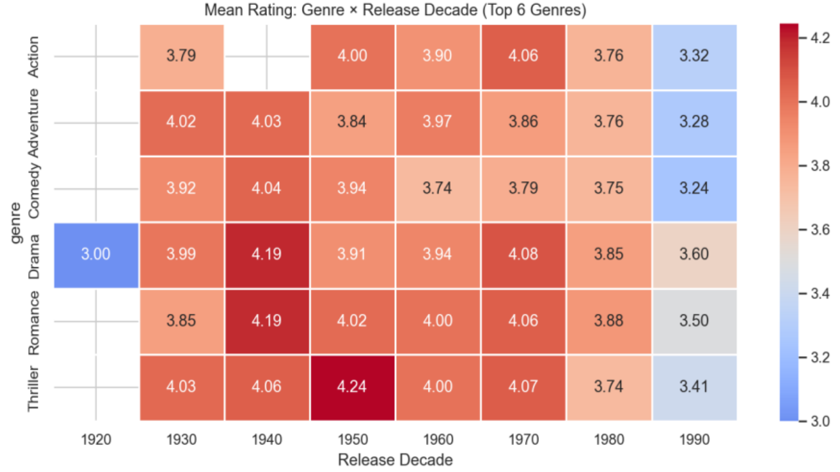
Consolidated relational schema entities (u.data, u.item, u.user, u.genre) to secure structural parity before training models.

- **Referential Integrity:** Validated indices, confirming zero missing entries across user-movie joins.
- **Parity Check:** Verified rating constraints ∈ [1, 5] bounds (0 records discarded).
- **Feature Augmentation:** Appended rating_date (POSIX to datetime) and release_year.
- **Final Asset:** Static exports in ml100k_clean.csv with minor missing values (<0.01% on release data,100,000 records × 12 engineered features)



Key EDA Findings (M4)





Rating Behavior & Demographics

- **Positive Skewness:** High score concentration exists. Stars (4 & 5) comprise 55.4% of ratings, while 1-star ranks only 6.1%.
- **Central Tendency:** Global baseline average stands high at **3.53**.
- **Demographic Skew:** Active reviewers center at 25 years old. 74% are male, and student jobs comprise 22% of total scores.

> **Key Hypothesis Shift:** Older cohort targets score more generously. Classic movies score consistently higher than modern 1990s counterparts due to survival/nostalgia bias. Gender proved to have a marginally weak effect on ratings...

Three Models, Two Evaluation Goals

ALGORITHM	TARGET SCOPE	MATHEMATICAL FORMULATION
1. Bias Baseline	Benchmark Lower Bound	$r_{ui} = \mu + b_u + b_i$
2. Item-Based CF	Localized Similarity	Cosine metric across k=30 neighbors
3. SVD Latents	Dimensional Reductions	$\hat{r}_{ui} = \mu + b_u + b_i + p_u^T q_i$

Evaluation Targets

We test against two separate criteria families using a held-out test split:

Group A (Rating Error): Root Mean Squared Error (RMSE) & Mean Absolute Error (MAE).

Group B (Discovery List): Precision@10 & Recall@10 (evaluating items with score ≥ 4).

Results on u1.test (20,000 ratings)

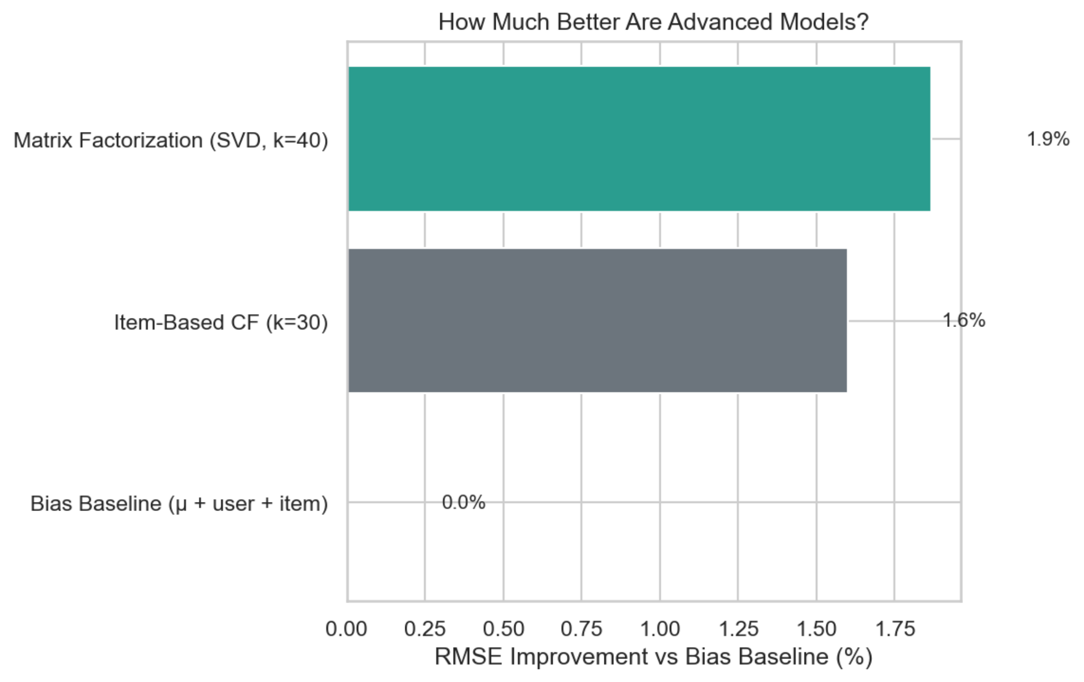
Model Framework	RMSE↓	MAE↓	P@10↑	R@10↑
Bias Baseline	0.977	0.766	0.009	0.003
Item-Based CF (k=30)	0.962	0.751	**0.062**	**0.014**
Matrix Factorization (k=40)	**0.959**	**0.750**	0.025	0.008

SVD Accuracy Win

Best RMSE score of **0.959** (-1.9% error over baseline), validating low-rank theory.

Item-CF List Win

Precision@10 of **6.2%** (2.5x increase over SVD) due to less conservative list mapping.



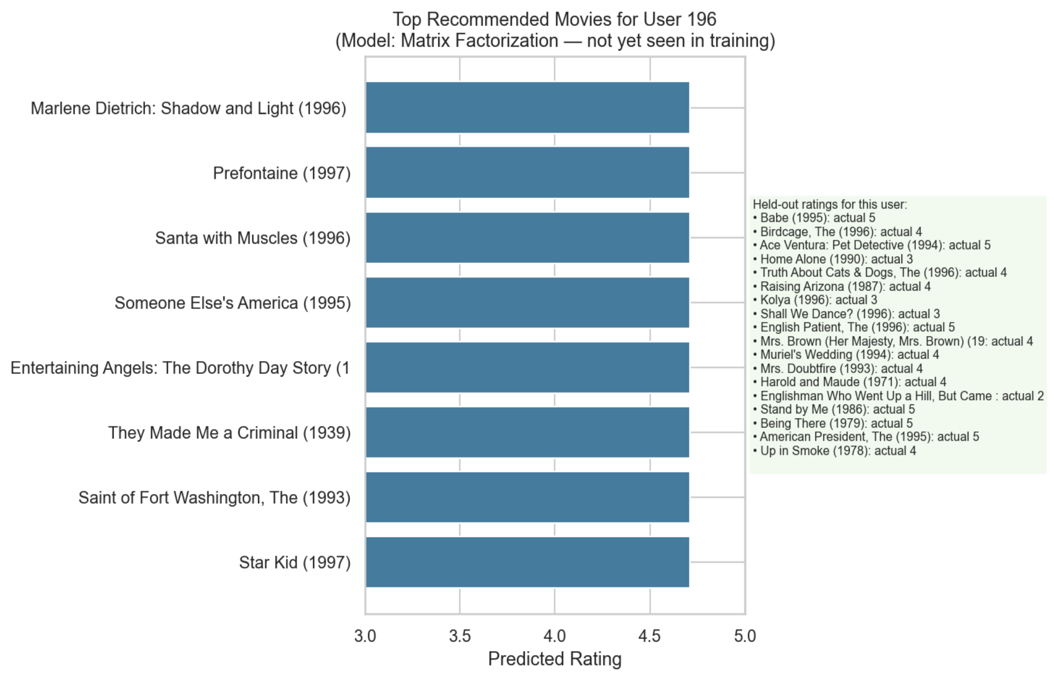
Prediction Behavior: All models exhibit conservative forecasting, compressing rating outputs toward the 3–4 star range.

From Metrics to Product Decisions

Deployment Playbook

Commercial Goal	Best Model Selection	Rationale
Minimize rating prediction error	Matrix Factorization	Lowest RMSE score
Maximize playlist conversions	Item-Based CF	Highest Precision@10 list score
System explainability	Bias Baseline	Auditable user & item offsets

Engineers must match deployments to localized catalog requirements instead of declaring a single system 'winner'.



What Can Go Wrong (Failure Modes)

COLD-START PROBLEM

New active profiles or fresh catalog entries without historical interaction indices bypass embeddings, forcing system fallbacks to global median values.

Solution: Popularity-based seeding is required during customer onboardings.

SYSTEM BIASES

Demographic Imbalance: High skew towards young males (74%) leads to weaker recommendation signals for other cohorts.

Over-compression: Conservative behavior clusters scores around 3–4 stars, rendering 1-star forecasts highly inaccurate.

Evaluation Limits: Single fixed split (u1); robust confirmation requires k-fold Cross-Validation.

Conclusion & Future Roadmap

- **End-to-End Pipeline:** Delivered clean, reproducible engineering logic spanning ingest, custom data exploration, modeling, and deployment guides.
- **SVD Strengths:** Latent structures reliably capture broad global ratings, while neighborhood architectures better preserve ranking metrics.
- **EDA Implications:** Recognizing historical skews remains vital to prevent propagating popularity feedback loops.

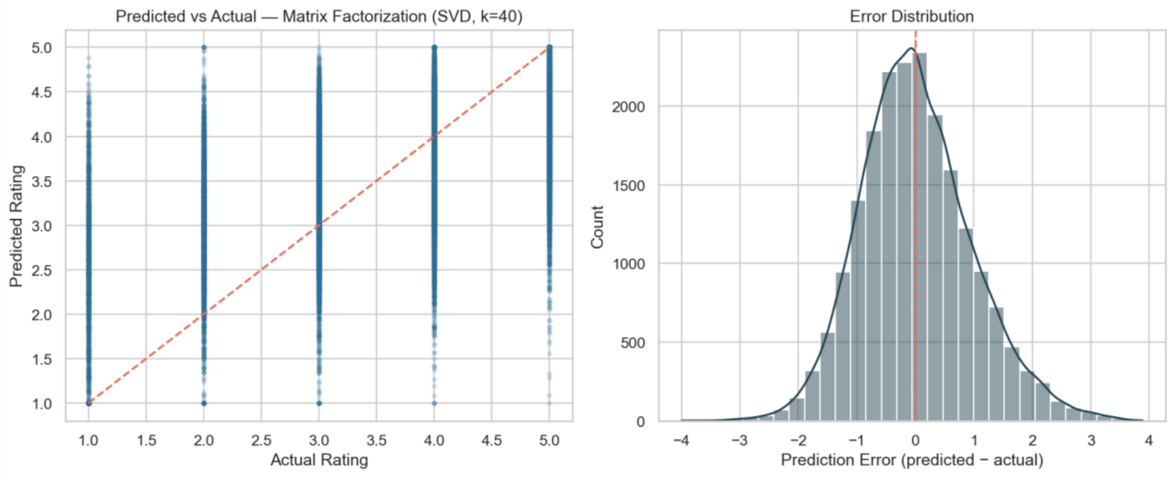
93.7%

SPARSITY SUCCESSFULLY RESOLVED

1. **Pipeline Ingest** — Cleaned raw tabular csv
2. **SVD Latents** — Optimized 40 latent factors
3. **Evaluation** — Identified failure boundaries
4. **Hybrid Model** — Incorporate genre features
5. **Fairness Audit** — Correct demographic skew

THANK YOU !

Model Diagnostics



Scientific Reference Card

Researcher	System Responsibility
李仕轩 **(Lead)**	Experimental setup & evaluative scoping
刘畅	Pipeline ingest & exploratory findings
冯博伦	Baseline development & similarity design
郭甲进	SVD tuning & hyperparameter selection

Project Repository: https://github.com/36712/Data-Science-Projects